

SAGUACHE, CO, USA

WMO: 722333

Lat: 38.097N

Lon: 106.169W

Elev: 7826

StdP: 10.99

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 03069

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.4	2.9	-10.7	4.1	15.0	-7.9	4.8	16.1	32.8	33.5	28.2	30.1	6.0	330	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.0	86.1	53.8	83.8	53.4	81.3	53.4	59.2	75.2	57.9	73.3	56.9	72.5	10.4	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.7	85.5	61.8	53.8	82.8	61.2	51.9	77.0	61.8	29.6	75.5	28.7	74.4	27.9	70.7	63.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.8	24.3	20.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	22.1	28.8	38.7	44.6	51.7	62.3	67.0	64.6	58.4	46.8	36.0	22.6		
	DBStd	16.68	7.71	7.81	6.98	6.28	6.67	5.29	3.78	3.80	4.67	6.49	7.48	9.73		
	HDD50	3462	864	593	353	179	62	1	0	0	3	135	421	851		
	HDD65	7323	1329	1013	815	612	412	111	22	55	202	565	871	1316		
	CDD50	1772	0	0	2	17	115	369	528	452	255	34	0	0		
	CDD65	161	0	0	0	0	0	29	85	42	5	0	0	0		
	CDH74	3293	0	0	0	3	99	941	1285	732	230	3	0	0		
	CDH80	773	0	0	0	0	7	243	355	150	18	0	0	0		
(14) Wind	WSAvg	8.2	6.9	8.7	9.8	10.2	9.1	9.0	7.5	7.0	6.6	8.5	8.4	7.4		
(15) Precipitation	PrecAvg	8.4	0.2	0.2	0.4	0.6	0.7	0.6	1.5	1.6	1.0	0.7	0.4	0.3		
	PrecMax	14.8	1.0	1.3	1.3	2.7	2.0	2.4	3.6	3.4	3.1	2.8	1.7	2.0		
	PrecMin	5.2	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.4	0.1	0.0	0.0	0.0		
	PrecStd	2.1	0.2	0.2	0.3	0.6	0.5	0.5	0.9	0.7	0.7	0.7	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	54.4	67.8	73.1	81.0	89.6	90.7	86.3	82.1	73.2	63.0	50.5		
		MCWB	33.7	37.3	42.5	44.7	48.8	53.1	54.6	54.1	51.7	48.4	41.7	36.3		
	2%	DB	42.9	51.5	63.4	68.0	77.1	85.8	86.3	83.9	79.2	70.0	58.9	45.4		
		MCWB	31.2	35.8	40.4	43.5	47.7	52.1	55.0	54.1	50.8	47.0	40.4	33.0		
	5%	DB	38.6	45.9	59.2	64.1	72.9	82.3	83.9	81.4	77.0	65.7	54.5	41.3		
		MCWB	30.1	33.8	38.7	41.5	46.2	51.0	55.0	54.7	51.2	45.1	38.2	31.1		
	10%	DB	34.5	42.7	54.4	61.2	69.9	80.6	81.3	78.9	73.2	62.8	50.4	36.8		
		MCWB	28.6	31.7	37.0	40.5	45.7	50.7	55.0	54.7	49.8	44.0	36.7	29.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.7	39.3	43.2	46.5	51.4	57.7	61.1	60.8	57.2	52.2	43.3	37.6		
		MCDB	43.6	51.6	61.9	66.1	70.0	74.4	78.1	74.9	74.1	65.9	58.8	45.2		
	2%	WB	33.2	36.8	41.2	44.9	49.7	55.7	59.4	59.1	55.3	49.2	41.2	34.7		
		MCDB	38.1	47.5	60.7	63.8	69.5	75.9	76.4	73.7	70.6	63.4	56.6	42.7		
	5%	WB	31.3	34.7	39.3	43.1	48.2	54.3	58.3	57.8	53.8	47.0	39.2	32.2		
		MCDB	37.3	44.6	58.1	62.3	67.7	74.9	75.2	71.5	68.9	61.5	53.6	39.9		
	10%	WB	29.2	32.6	37.5	41.5	46.7	52.6	57.2	56.9	52.4	45.0	37.0	30.1		
		MCDB	34.9	41.1	52.4	58.7	65.9	73.4	73.9	70.7	67.8	60.1	49.1	37.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.4	23.3	26.8	28.1	28.9	32.8	30.0	28.4	30.7	28.2	25.7	22.6		
		MCDBR	24.6	26.5	33.1	34.7	34.8	36.4	33.8	31.9	33.6	33.4	30.4	25.9		
		MCWBR	16.4	15.2	16.1	16.2	15.2	15.3	12.0	11.7	14.3	16.3	16.6	16.2		
	5% WB	MCDBR	20.8	24.3	30.9	31.9	31.2	31.5	28.5	26.4	28.6	29.5	28.8	23.9		
		MCWBR	14.8	14.7	15.4	15.3	14.0	13.5	10.6	10.4	12.6	15.0	16.0	15.7		
(40) Clear-Sky Solar Irradiance	taub		0.223	0.229	0.257	0.280	0.299	0.304	0.348	0.330	0.288	0.260	0.235	0.226		
	taud		2.574	2.573	2.537	2.507	2.510	2.502	2.434	2.459	2.589	2.637	2.625	2.567		
	Ebn at Noon		317	326	323	318	311	308	293	297	307	310	309	308		
	Edh at Noon		22	26	30	33	33	34	36	34	28	24	21	21		
(44) All-Sky Solar Radiation	RadAvg		922	1232	1639	1947	2142	2359	2049	1872	1705	1336	992	799		
	RadStd		48	72	116	112	161	137	111	87	97	88	59	46		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	+43

Nomenclature: See separate page